

A Partial Van Corput Reduction for Four-Term Progression-Free Sets

Partial-results manuscript

June 2026

Abstract

Let $r_k(N)$ denote the maximum size of a subset of $\{1, \dots, N\}$ with no non-trivial k -term arithmetic progression. We record a partial van Corput-type reduction from the $k = 4$ problem to the $k = 3$ problem. For a 4-term-progression-free set $A \subseteq \{1, \dots, N\}$, write

$$A_d = \{x \in A : x + d \in A\}.$$

We prove that there is a popular difference d satisfying

$$|A_d| \geq |A|^2/(4N),$$

and that such a fiber avoids 3-term progressions of step d and $2d$. The missing ingredient for a full reduction is the assertion that some popular fiber is free of all 3-term progressions. We isolate this as Conjecture 2. Assuming Conjecture 2, one obtains

$$r_4(N) \leq 2\sqrt{Nr_3(N)}.$$

Combining this conditional reduction with Raghavan's 2026 bound for $r_3(N)$ would give a conditional quasi-polynomial upper bound for $r_4(N)$. We also describe a 2×3 grid reformulation of the gap, Lean 4 formalization status, and computational evidence supporting the conjecture.

Contents

1	Introduction	2
2	Setup and Difference Fibers	2
3	What 4-AP-Freeness Forces	3
4	The Exact Gap	4
5	Conditional Reduction from r_4 to r_3	5
6	A 2×3 Grid Reformulation	6
7	Computational Evidence	6
7.1	Exhaustive small- N checks	6
7.2	Random greedy sampling	6
7.3	Observed structural patterns	7

8	A Refined Empirical Conjecture	7
9	Lean 4 Formalization	7
9.1	Formalized definitions	8
9.2	Formalized partial van Corput structure	8
9.3	Declarations still using <code>sorry</code>	8
10	What Is Actually Being Submitted	8
11	Open Problems	9

1 Introduction

For $k \geq 2$, let $r_k(N)$ be the largest size of a subset

$$A \subseteq \{1, \dots, N\}$$

containing no non-trivial k -term arithmetic progression. Thus A is k -AP-free if it contains no set

$$\{a, a + d, \dots, a + (k - 1)d\}$$

with $d > 0$.

Szemerédi's theorem gives $r_k(N) = o(N)$ for each fixed $k \geq 3$, but the quantitative behavior remains difficult. The $k = 3$ case has seen major recent progress. The input used in this note is Raghavan's 2026 bound:

$$r_3(N) \leq C_1 N \exp\left(-c_1 \frac{(\log N)^{1/6}}{\log \log N}\right)$$

for absolute constants $C_1, c_1 > 0$ and all sufficiently large N .

For $k = 4$, the best unconditional published upper bounds are much weaker than this quasi-polynomial scale. The purpose of this note is not to claim an unconditional improvement, but to isolate a precise combinatorial condition under which Raghavan's $r_3(N)$ bound transfers to $r_4(N)$. The transfer rests on popular difference fibers.

2 Setup and Difference Fibers

Let

$$A \subseteq \{1, \dots, N\}, \quad |A| = M.$$

For $1 \leq d \leq N - 1$, define the difference fiber

$$A_d = \{x \in A : x + d \in A\}.$$

Equivalently, $A_d = A \cap (A - d)$, written inside $\{1, \dots, N\}$.

The first observation is a pigeonhole count. It is elementary, but it is the starting point of the whole reduction.

Lemma 2.1 (Popular difference). *Let $A \subseteq \{1, \dots, N\}$ have size $M \geq 2$. Then there exists $d \in \{1, \dots, N - 1\}$ such that*

$$|A_d| \geq \frac{M^2}{4N}.$$

Proof. Count ordered pairs $(a, b) \in A^2$ with $a < b$. Every such pair determines a unique positive difference $d = b - a$, and contributes one element a to A_d . Therefore

$$\sum_{d=1}^{N-1} |A_d| = \binom{M}{2} = \frac{M(M-1)}{2}.$$

Since $M \geq 2$,

$$\frac{M(M-1)}{2} \geq \frac{M^2}{4}.$$

There are at most $N-1 < N$ possible positive differences. Hence some d satisfies

$$|A_d| \geq \frac{M^2/4}{N} = \frac{M^2}{4N}.$$

□

We call a difference d *popular* if

$$|A_d| \geq \frac{M^2}{4N}.$$

Let

$$\mathcal{S}(A) = \left\{ d \in \{1, \dots, N-1\} : |A_d| \geq \frac{M^2}{4N} \right\}$$

be the set of popular differences.

3 What 4-AP-Freeness Forces

Suppose now that A is 4-AP-free. The key unconditional structural fact is that fibers A_d avoid certain specific 3-term progressions.

Lemma 3.1 (Step- d freeness). *Let $A \subseteq \{1, \dots, N\}$ be 4-AP-free and let $d > 0$. Then A_d contains no 3-term arithmetic progression with common difference d . In other words, there is no x such that*

$$x, x+d, x+2d \in A_d.$$

Proof. Assume for contradiction that $x, x+d, x+2d \in A_d$. By definition of A_d ,

$$x \in A_d \Rightarrow x \in A, x+d \in A,$$

$$x+d \in A_d \Rightarrow x+d \in A, x+2d \in A,$$

and

$$x+2d \in A_d \Rightarrow x+2d \in A, x+3d \in A.$$

Thus

$$x, x+d, x+2d, x+3d \in A,$$

which is a non-trivial 4-term arithmetic progression in A . This contradicts the assumption that A is 4-AP-free. □

Lemma 3.2 (Step- $2d$ freeness). *Let $A \subseteq \{1, \dots, N\}$ be 4-AP-free and let $d > 0$. Then A_d contains no 3-term arithmetic progression with common difference $2d$. In other words, there is no x such that*

$$x, x+2d, x+4d \in A_d.$$

Proof. Assume for contradiction that $x, x + 2d, x + 4d \in A_d$. From $x \in A_d$, we get

$$x \in A, \quad x + d \in A.$$

From $x + 2d \in A_d$, we get

$$x + 2d \in A, \quad x + 3d \in A.$$

Therefore

$$x, x + d, x + 2d, x + 3d \in A,$$

again a 4-term arithmetic progression in A . This is impossible. $\square$

Combining Lemmas 2.1, 3.1, and 3.2 gives the main unconditional partial result.

Proposition 3.3 (Partial van Corput fiber structure). *Let $A \subseteq \{1, \dots, N\}$ be 4-AP-free and let $|A| = M \geq 2$. Then there exists $d \in \{1, \dots, N - 1\}$ such that:*

1. $|A_d| \geq M^2/(4N)$;
2. A_d contains no 3-term arithmetic progression with common difference d ;
3. A_d contains no 3-term arithmetic progression with common difference $2d$.

Proof. Choose d by Lemma 2.1. Since A is 4-AP-free, Lemma 3.1 gives property (2), and Lemma 3.2 gives property (3). $\square$

4 The Exact Gap

Proposition 3.3 is not enough to apply $r_3(N)$, because $r_3(N)$ bounds sets with no 3-term arithmetic progressions of any positive common difference. The proposition only rules out common differences d and $2d$ inside A_d .

This distinction is essential.

Remark 4.1 (A warning example). *Let*

$$A = \{1, 2, 3, 5, 6, 8, 9\} \subseteq \{1, \dots, 9\}.$$

This set is 4-AP-free. For $d = 1$,

$$A_1 = \{1, 2, 5, 8\}.$$

But A_1 contains the 3-term progression

$$2, 5, 8,$$

whose common difference is 3. Thus even a natural fiber of a 4-AP-free set need not be fully 3-AP-free.

The missing assertion is the following.

Conjecture 4.2 (Conjecture 2: popular fully 3-AP-free fiber). *Let $A \subseteq \{1, \dots, N\}$ be 4-AP-free, with $|A| = M \geq 2$. Then there exists $d \in \mathcal{S}(A)$ such that A_d is fully 3-AP-free.*

Equivalently, among the popular differences, at least one fiber has no non-trivial 3-term progression with any positive common difference.

5 Conditional Reduction from r_4 to r_3

Conjecture 4.2 gives the desired reduction immediately.

Theorem 5.1 (Conditional van Corput reduction). *Assume Conjecture 4.2. Then for every N ,*

$$r_4(N) \leq 2\sqrt{Nr_3(N)}.$$

Proof. Let $A \subseteq \{1, \dots, N\}$ be 4-AP-free and set $M = |A|$. If $M < 2$, the claim is trivial. Assume $M \geq 2$.

By Conjecture 4.2, there exists a popular d such that A_d is fully 3-AP-free. Since d is popular,

$$|A_d| \geq \frac{M^2}{4N}.$$

Since $A_d \subseteq \{1, \dots, N\}$ and is 3-AP-free,

$$|A_d| \leq r_3(N).$$

Therefore

$$\frac{M^2}{4N} \leq r_3(N),$$

so

$$M \leq 2\sqrt{Nr_3(N)}.$$

Taking A to have maximum possible size $M = r_4(N)$ gives the result. $\square$

Corollary 5.2 (Conditional quasi-polynomial r_4 bound). *Assume Conjecture 4.2. Assume also Raghavan's bound*

$$r_3(N) \leq C_1 N \exp\left(-c_1 \frac{(\log N)^{1/6}}{\log \log N}\right)$$

for all sufficiently large N . Then there exist constants $C, c > 0$ such that for all sufficiently large N ,

$$r_4(N) \leq CN \exp\left(-c \frac{(\log N)^{1/6}}{\log \log N}\right).$$

Proof. By Theorem 5.1,

$$r_4(N) \leq 2\sqrt{Nr_3(N)}.$$

Using Raghavan's bound,

$$r_4(N) \leq 2\sqrt{N \cdot C_1 N \exp\left(-c_1 \frac{(\log N)^{1/6}}{\log \log N}\right)}.$$

Thus

$$r_4(N) \leq 2\sqrt{C_1} N \exp\left(-\frac{c_1}{2} \frac{(\log N)^{1/6}}{\log \log N}\right).$$

Set $C = 2\sqrt{C_1}$ and $c = c_1/2$. $\square$

Remark 5.3. *The square root changes the constant c_1 to $c_1/2$, but it does not change the exponent $1/6$ on $\log N$.*

6 A 2×3 Grid Reformulation

The remaining obstruction can be expressed as a grid-avoidance condition. A 3-term progression in A_d with step e means

$$x, x + e, x + 2e \in A_d.$$

By definition of A_d , this is equivalent to the six points

$$x, x + e, x + 2e, x + d, x + e + d, x + 2e + d$$

all lying in A . These six points form a 2×3 arithmetic grid:

$$\{x + ie + jd : i = 0, 1, 2, j = 0, 1\}.$$

The cases $e = d$ and $e = 2d$ are already excluded by Lemmas 3.1 and 3.2. Thus the unresolved part of Conjecture 4.2 is the possibility of such grids with

$$e \neq d, \quad e \neq 2d.$$

Define $G(A, d, N)$ to be the number of pairs (x, e) such that:

- $e > 0$;
- $e \neq d$ and $e \neq 2d$;
- $x, x + e, x + 2e \in A_d$.

Then Conjecture 4.2 is equivalent to:

$$\exists d \in \mathcal{S}(A) \quad G(A, d, N) = 0.$$

This is often the most useful formulation of the problem. It says that among popular differences, at least one column spacing d avoids all non-trivial 2×3 grids.

7 Computational Evidence

The repository contains several verification scripts and reports in `runs/erdos142/`. The core test is:

Given a 4-AP-free set $A \subseteq \{1, \dots, N\}$, compute all popular differences $d \in \mathcal{S}(A)$, compute each fiber A_d , and check whether at least one popular fiber is fully 3-AP-free.

7.1 Exhaustive small- N checks

The script `conjecture2_verify.py` exhaustively checks all 4-AP-free sets for small N . The project reports exhaustive verification through $N \leq 20$, covering approximately 478,000 4-AP-free sets, with no counterexample to Conjecture 4.2.

7.2 Random greedy sampling

For larger N , the project uses random greedy constructions of 4-AP-free sets. The reports state that random sampling was extended up to $N = 200000$, with no counterexample found across approximately 4893 trials.

This is evidence, not proof. Random greedy sets do not represent all possible 4-AP-free sets. Still, the experiments are useful because they reveal which naive strengthening attempts are false and where good fibers tend to occur.

7.3 Observed structural patterns

The computations show:

- Conjecture 4.2 holds in all tested examples.
- The largest fiber, $d = \arg \max |A_d|$, is often not 3-AP-free.
- Good fibers usually occur near the lower end of the popular-fiber size spectrum.
- The fraction of bad popular fibers can be very large; reported values reach approximately 0.9890 in the largest experiments.
- Despite the high bad fraction, the absolute number of good popular fibers grows in the sampled data.

8 A Refined Empirical Conjecture

Let

$$L = \frac{M^2}{4N}$$

be the popularity threshold and

$$P = \max_{d \in \mathcal{S}(A)} |A_d|$$

be the largest popular fiber size. For $d \in \mathcal{S}(A)$, define the witness fraction

$$\text{wfrac}(d) = \frac{|A_d|}{P}.$$

The computations suggest that good fibers often occur close to the popularity threshold L , and that in random greedy data

$$\frac{L}{P} \approx 0.20.$$

This motivates the following stronger empirical statement.

Conjecture 8.1 (Refined Conjecture 2: lower-quarter witness). *Let $A \subseteq \{1, \dots, N\}$ be 4-AP-free. Then there exists $d \in \mathcal{S}(A)$ such that A_d is fully 3-AP-free and*

$$|A_d| \leq \frac{1}{4} \max_{d' \in \mathcal{S}(A)} |A_{d'}|.$$

This refined conjecture is not needed for Theorem 5.1, but it may be easier to attack structurally because it points toward small popular fibers as the likely witnesses.

9 Lean 4 Formalization

The project includes Lean 4 files in the `lean/` directory. The purpose of the formalization is to check the finite combinatorial bookkeeping and to make the gap precise.

9.1 Formalized definitions

The file `ArithmeticProgressions.lean` defines:

- k -term arithmetic progressions as finite sets of the form

$$\{a + id : 0 \leq i < k\};$$

- the predicate `HasKAP A k`;
- basic lemmas about the cardinality of such progressions;
- a bridge between `HasKAP A 3` and Mathlib’s `ThreeAPFree`.

The file `RkN.lean` defines $r_k(N)$ as a maximum over finite subsets of $\{0, \dots, N - 1\}$, using Lean’s `Nat.findGreatest`. It also proves the bridge

$$r_3(N) = \text{rothNumberNat}(N),$$

linking the custom definition to Mathlib’s Roth-number definition.

9.2 Formalized partial van Corput structure

The file `VanCorput.lean` defines the fiber

$$A_d = \{x \in A : x + d \in A\}$$

as `fiberAtDiff A d`. It proves the Lean analogues of Lemmas 3.1 and 3.2. It also proves the popular-difference pigeonhole lemma and packages the result as a Lean theorem called `proposition_1`.

Thus Proposition 3.3 is genuinely formalized.

9.3 Declarations still using sorry

The important unfinished declarations are:

- `van_corput_inequality`, corresponding to $r_4(N) \leq 2\sqrt{Nr_3(N)}$;
- `rk_four_raghavan_bound`, corresponding to the conditional quasi-polynomial $r_4(N)$ bound;
- general Szemerédi statements for $k \geq 4$, because they are not available in Mathlib at the required level.

This is the correct formal status: the partial combinatorial result is proved, while the full reduction remains blocked by Conjecture 4.2.

10 What Is Actually Being Submitted

The safe partial result is Proposition 3.3, together with the isolation of Conjecture 4.2 and the conditional consequence Theorem 5.1.

The submission should not claim an unconditional new upper bound for $r_4(N)$. The correct claim is:

We prove that every 4-AP-free set has a popular difference fiber satisfying a large lower bound and avoiding 3-APs of steps d and $2d$. We isolate the remaining full-freeness condition as Conjecture 2. Conditional on this conjecture, Raghavan’s $r_3(N)$ bound implies a quasi-polynomial upper bound for $r_4(N)$. Computational evidence supports the conjecture.

This is a meaningful partial result because it:

1. gives a clean and formally checked decomposition of the attempted van Corput reduction;
2. identifies the exact obstruction rather than hiding it in an incorrect “fiber is 3-AP-free” claim;
3. gives a concrete finite conjecture that can be tested computationally;
4. reformulates the obstruction as a 2×3 grid-avoidance problem.

11 Open Problems

1. Prove or disprove Conjecture 4.2.
2. Prove a weaker statement still sufficient for $r_4(N) \leq C\sqrt{Nr_3(N)}$.
3. Prove a positive lower bound on the number of good popular fibers, or show that at least one must exist by an averaging or energy argument.
4. Establish structural restrictions on the 2×3 grids counted by $G(A, d, N)$.
5. Test Conjecture 4.2 on adversarial constructions beyond random greedy 4-AP-free sets.
6. Formalize the equivalence $G(A, d, N) = 0$ iff A_d is fully 3-AP-free, modulo the already-proved step- d and step- $2d$ exclusions.

References

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